

WASP-36b

R-0743

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_250313 · created 2026-07-28T02:05:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460747.6752 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.047
Transit depth (Rp/Rs)^2	2.25 +/- 1.42 [%]
Orbital Inclination (inc)	87.45 +/- 2.93
Airmass coefficient 1 (a1)	42.14 +/- 29.51
Airmass coefficient 2 (a2)	-2.99 +/- 1.51
Scatter in the residuals of the lightcurve fit is	56.3 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	3.0
Transit Duration (day)	0.0239 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.15 $\pm$ 0.047 vs 0.1368 (deviation 0.28 σ)
Duration fit vs published	0.0239 d vs 0.0758 d (deviation 6.74 σ)
Mid-time O-C	-14.4 min
Depth SNR	3.2
Residual scatter	56.3 %

Light curve

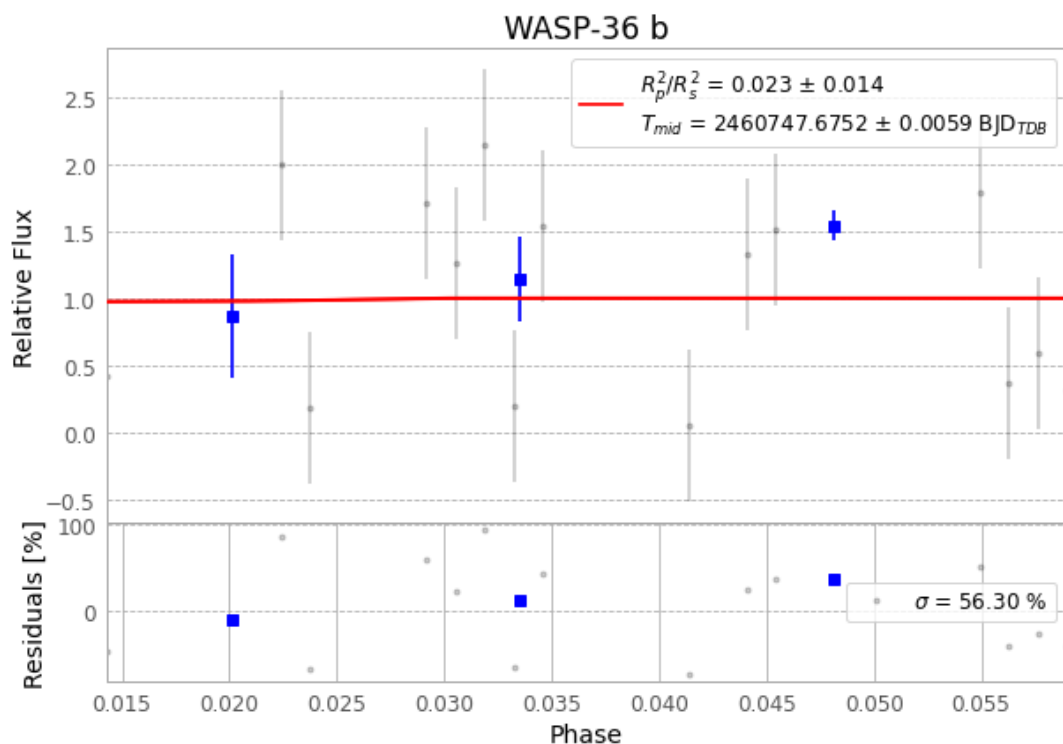


Figure 1 — FinalLightCurve_WASP-36 b_2025-03-12.png (SHA-256 951377a9671da199...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [517, 430], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1aca773e83f68de6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

35 FITS frames (23 MB), WASP-36250313043315.FITS → WASP-36250313061515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-250313.log (c94412e1a8c0...), FinalLightCurve_WASP-36 b_2025-03-12.png (951377a9671d...), FinalLightCurve_WASP-36 b_2025-03-12.csv (4cb7b04419ed...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:05Z.